END-END COMMUNICATION AT TRANSPORT LAYER

AIM: To Implement echo client server using TCP/UDP sockets.

TCP:

CODE:-

```
import socket
import threading
def start_server():
    server = socket.socket(socket.AF_INET,
    server.bind(("127.0.0.1", 5000))
    server.listen()
    print("[SERVER] Running on 127.0.0.1:2000")
    while True:
        data, addr = s.recvfrom(1024)
        msg = data.decode()
        if msg.lower() == "quit":
            print(f"[SERVER] Client {addr}
                  ended the connection.")
            break
        print(f"[SERVER] Received from {addr}: {msg}")
        server.sendto(data, addr)
    server.close()
    print("[SERVER] Stopped")

def start_client():
    client = socket.socket(socket.AF_INET, socket.
                           SOCK_STREAM)
    client.connect(("127.0.0.1", 5000))
    print("[CLIENT] connected to server")
    while True:
        msg = input()
        client.send(msg.encode())
        if msg.lower() == "quit":
            break
        response = client.recv(1024).decode()
        print(f"[Server] {response}")
```

UDP:

Code:

```
import socket
import threading
def udp_server():
    s = socket.socket(
            socket.AF_INET
            socket.SOCK_DGRAM
        )
    s.bind(("127.0.0.1", 12000))
    print("[SERVER] Running on 127.0.0.1:12000")

def udp client():
    c = socket.socket(
            socket.AF_INET,
            socket.SOCK_DGRAM,
        )
    server = ('127.0.0.1', 12000)
    while True:
        msg = input("Enter message (or 'quit'): ")
        c.sendto(msg.encode(), server)
        if msg.lower() == 'quit':
            print("[CLIENT] closing connection")
            break
        data, _ = c.recvfrom(1024)
        print("Echo from server: ", data.decode())
    c.close()
    print("[CLIENT] stopped.")
```

RESULT:

Thus end to end communication is implemented successfully at transport